

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S1 (S,FE) S2 (S,FE) Examination May 2025 (2019 Scheme)

Course Code: MAT101**Course Name: LINEAR ALGEBRA AND CALCULUS****(2019-Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

1 Find the rank of the matrix $\begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$ (3)

2 Write down the matrix of the quadratic form $2x^2 + 5y^2 - 6z^2 - 2xy - yz + 8xz$ (3)

3 If $w = \tan^{-1}(xyz)$, find the differential dw at $(1,1,1)$. (3)

4 Find the slope of the function $f(x,y) = y\cos(xy) + \sin(xy)$ at $(\pi,1)$ along the x - direction. (3)

5 Evaluate $\int_{-1}^2 \int_0^2 \int_0^1 (x^2 + y^2 + z^2) dx dy dz$ (3)

6 $\iint_R (x \sin y - y \sin x) dA$ over the region $R = \{(x,y), 0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{2}\}$ (3)

7 Test the convergence of the series $\sum_{k=1}^{\infty} \frac{2k+1}{(k+1)^2}$ (3)

8 Check the convergence of the series $\sum_{k=1}^{\infty} \frac{k}{2^k}$ (3)

9 Find the Taylors series for $f(x) = \sin \pi x$ about $x = \frac{1}{2}$ upto fourth degree (3)

term.

- 10 Find the Fourier constant a_n for the function $f(x) = x^2$ in the interval $(-\pi, \pi)$. (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Solve the following linear system of equations using Gauss elimination method. (7)

$$4y + 4z = 24$$

$$3x - 11y - 2z = -6$$

$$6x - 17y + z = 18$$

- b) Find the eigenvalues and eigenvectors of (7)

$$\begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

- 12 a) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 2 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix}. \text{ Also find the diagonal matrix.}$$

- b) Find the canonical form corresponding to the quadratic form (7)

$$8x^2 + 7y^2 + 3z^2 - 12xy - 8yz + 4xz. \text{ Also find the definiteness.}$$

Module-II

- 13 a) Find the local linear approximation L to the function $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$ at (7)

the point P(4, 3). Compare the error in approximating f by L at the point Q(3.92, 3.01).

- b) Find the absolute maximum and minimum values of (7)

$f(x, y) = 3xy - 6x - 3y + 7$ on the closed triangular region R with vertices $(0,0), (3,0), (0,5)$.

14 a) If $w = r^2 - r \tan \theta$, $r = \sqrt{s}$, $\theta = \pi s$ at $s = \frac{1}{4}$, evaluate $\frac{dw}{ds}$ at $s = \frac{1}{4}$. (7)

The focal length of a mirror is given by $\frac{2}{f} = \frac{1}{v} - \frac{1}{u}$. Use differentials to (7)

- b) find the percentage error in f if u and v are both in error by 2% each.

Module-III

15 a) Change the order of integration and hence evaluate $\int_0^4 \int_y^4 \frac{x}{x^2 + y^2} dx dy$ (7)

- b) Find the total mass and center of gravity of the lamina with density $\delta(x, y) = xy$ in the first quadrant bounded by the circle $x^2 + y^2 = 1$ and the coordinate axes. (7)

16 a) Use double integral to find the volume of the solid bounded by the cylinder $x^2 + y^2 = 16$ and the planes $z = 0$, $z = 4 - x$ (7)

- b) Evaluate $\iint_R y dx dy$ over the positive quadrant of the circle $x^2 + y^2 = 25$ and the line $x + y = 5$. (7)

Module-IV

17 a) Find the general term of the series $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{1 \cdot 3 \cdot 5 \cdot 7} + \dots$ and use (7)

the ratio test to show that the series converges.

b) Test the absolute or conditional convergence of $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k^2}{k^3 + 1}$ (7)

18 a) Discuss the convergence of $\sum_{k=1}^{\infty} \frac{x^k}{k^3 + 1}$ (7)

b) Verify the convergence of the series $\sum_{k=0}^{\infty} \left(\frac{5^k + 1}{3^k} \right)$ (7)

Module-V

- 19 a) Expand the function as a Fourier series (7)
$$f(x) = x - x^2, \quad -1 < x < 1$$
- b) Find the half range cosine series for $f(x) = x(\pi - x)$, $0 < x < \pi$. (7)
- 20 a) Find the Fourier series of f defined by $f(x) = \begin{cases} 1+x, & -\pi < x < 0 \\ 1-x, & 0 < x < \pi \end{cases}$ (7)
- b) Obtain Fourier sine series for the function $f(x) = \cos x$, $0 < x < \pi$ (7)
